Task 1: Data Preprocessing and Handling Missing Values

In [5]:

```python
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.preprocessing import LabelEncoder, OneHotEncoder
from sklearn.feature_selection import SelectKBest, chi2
from sklearn.linear_model import LogisticRegression
from sklearn.feature_selection import RFE
from sklearn.ensemble import RandomForestClassifier
from sklearn.decomposition import PCA

data = pd.read_csv('Titanic-Dataset.csv')
data.head()

missing_values = data.isnull().sum()
print(missing_values)

data['Age'] = data['Age'].fillna(data['Age'].median())

data['Cabin'] = data['Cabin'].fillna("U")
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])

data.drop(columns=['Ticket', 'Cabin'], inplace=True)
data.isnull().sum()
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Out[5]:
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Fare           0
Embarked       0
dtype: int64
```

Task 2 Encoding categorical variables

In [6]:
```
categorical_cols = ['Sex', 'Embarked']

label_encoder = LabelEncoder()
data['Sex'] = label_encoder.fit_transform(data['Sex'])

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

data.head()
```

Out[6]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Fare	Embarked_Q	Embarked_S
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	7.2500	0	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	0	38.0	1	0	71.2833	0	0
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	7.9250	0	1
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	53.1000	0	1
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	8.0500	0	1

Task 3: Creating New Features

In [7]:

```
data['FamilySize'] = data['SibSp'] + data['Parch'] + 1

data['IsAlone'] = (data['FamilySize'] == 1).astype(int)

data[['FamilySize', 'IsAlone']].head()
```

Out[7]:

	FamilySize	IsAlone
0	2	0
1	2	0
2	1	1
3	2	0
4	1	1

Task 4: Feature Scaling

In [8]:

```
numerical_cols = ['Age', 'Fare', 'FamilySize']
scaler = StandardScaler()
data[numerical_cols] = scaler.fit_transform(data[numerical_cols])

data[numerical_cols].head()
```

Out[8]:

	Age	Fare	FamilySize
0	-0.565736	-0.502445	0.059160
1	0.663861	0.786845	0.059160
2	-0.258337	-0.488854	-0.560975
3	0.433312	0.420730	0.059160
4	0.433312	-0.486337	-0.560975

Task 5: Feature Selection Mechanisms

1. Univariate Selection using Chi-Square Test

In [9]:
```
X = data.drop(columns=['Survived', 'PassengerId', 'Name'])
y = data['Survived']
X = X.select_dtypes(include=['int64', 'uint8', 'float64']).clip(lower=0)
chi2_selector = SelectKBest(score_func=chi2, k=5)
X_chi2 = chi2_selector.fit_transform(X, y)
chi2_selector.scores_
```

Out[9]:
```
array([3.08736994e+01, 1.03041776e-01, 2.58186538e+00, 1.00974991e+01,
       1.17233400e+02, 1.08467891e-02, 5.48920482e+00, 4.04685858e+00])
```

Recursive Feature Elimination (RFE)

In [10]:
```
model = LogisticRegression()
rfe = RFE(estimator=model, n_features_to_select=5)
X_rfe = rfe.fit_transform(X, y)
rfe.support_
```

Out[10]:
```
array([ True, False,  True,  True, False, False,  True,  True])
```

3. Feature Importance using Random Forest

In [12]:
```
rf_model = RandomForestClassifier()
rf_model.fit(X, y)
importance = rf_model.feature_importances_
importance
```

Out[12]:
```
array([0.16377115, 0.32490158, 0.05936245, 0.06247596, 0.22563489,
       0.02540808, 0.04259089, 0.09585501])
```

Task 6: Dimensionality Reduction with PCA

In [13]:

```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)
pca.explained_variance_ratio_
```

Out[13]:

```
array([0.43561895, 0.23907554])
```